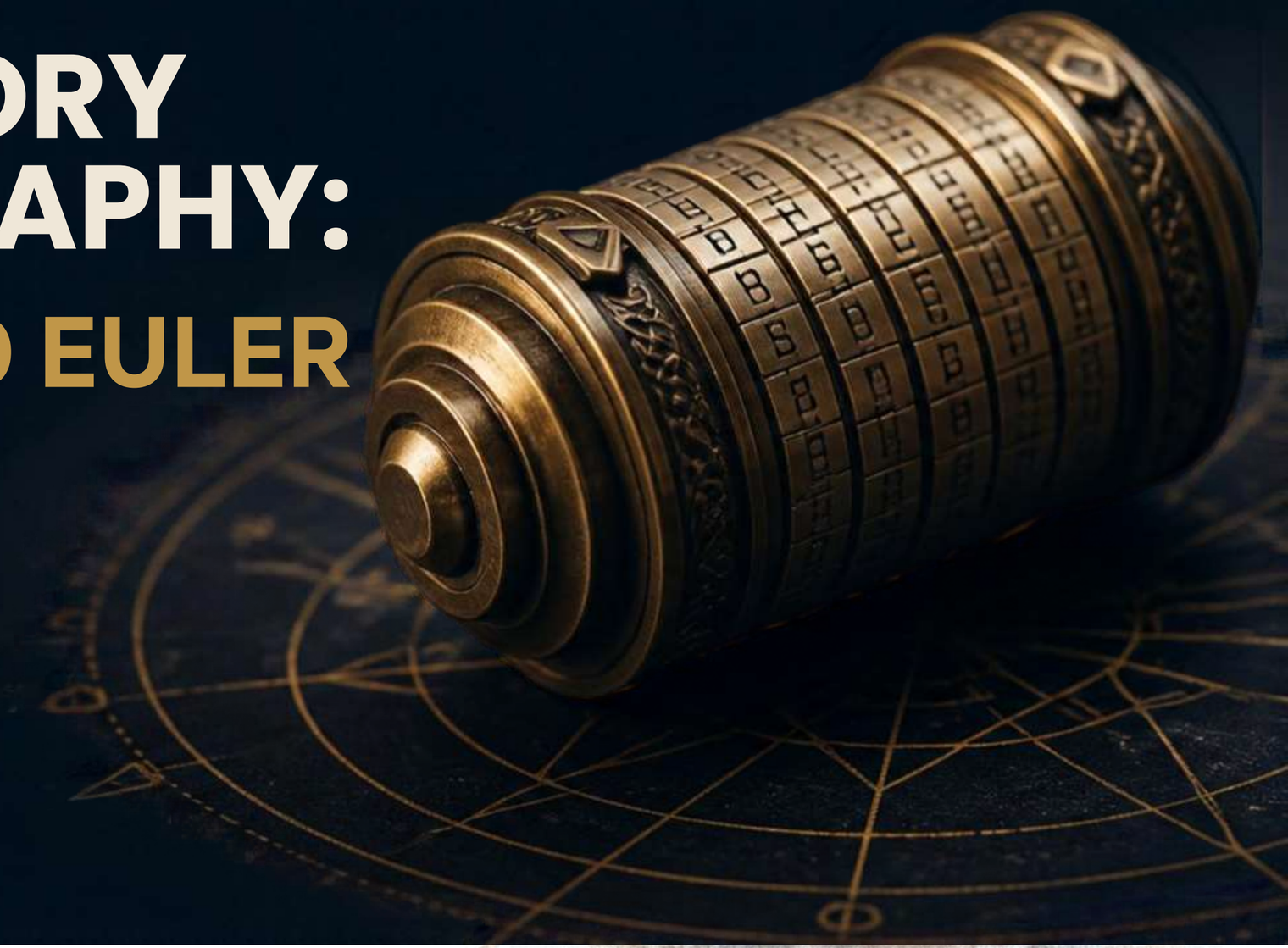
ROLE OF NUMBER THEORY IN CRYPTOGRAPHY:

FROM EUCLID TO EULER



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HISTORICAL FOUNDATIONS OF NUMBER THEORY IN CRYPTOGRAPHY

Number theory forms the mathematical backbone of cryptography

Ideas shaped by Euclid and Euler

Centuries of insight securing modern data



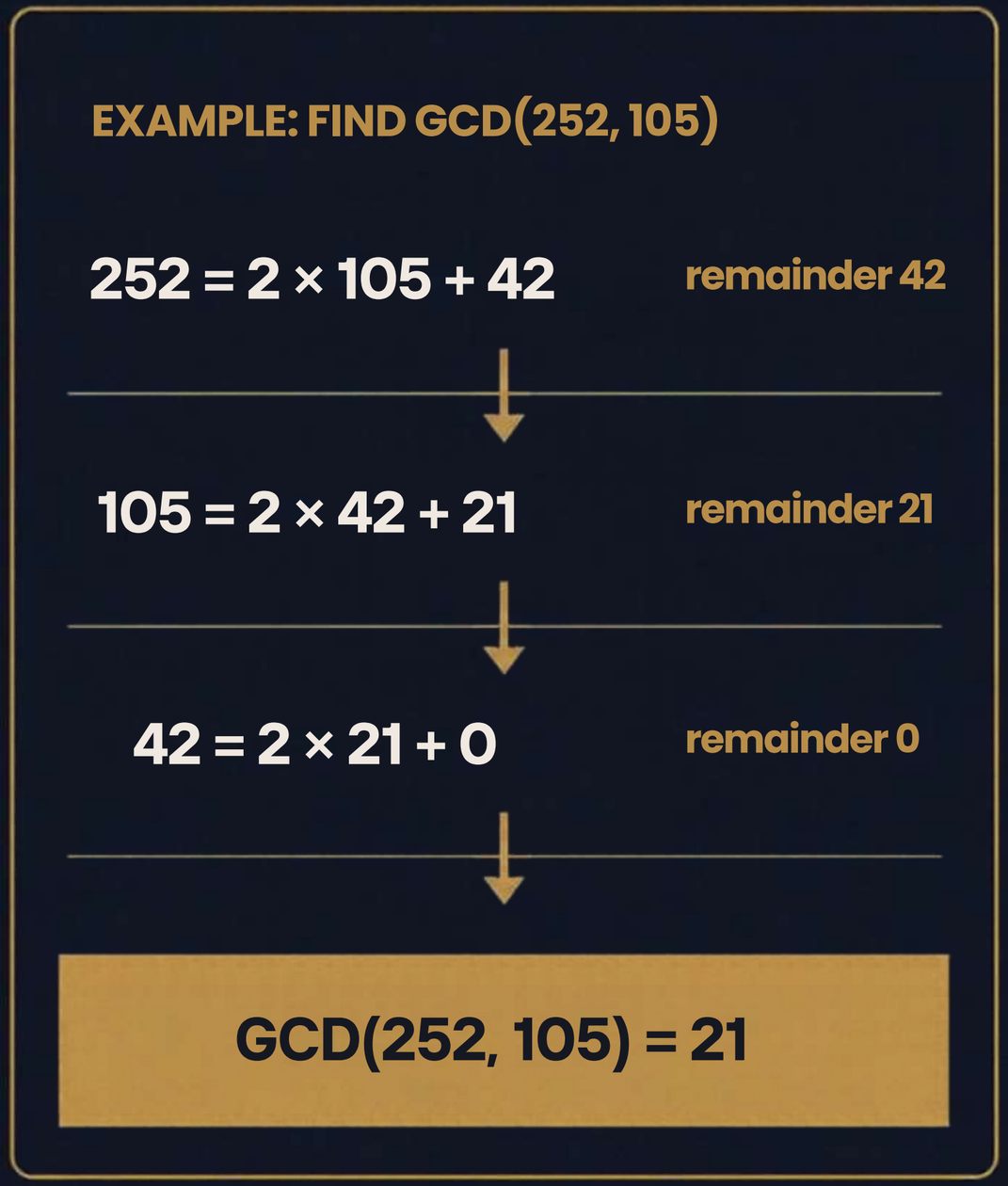
EUCLID'S ALGORITHM AND ITS CRYPTOGRAPHIC SIGNIFICANCE

Euclid's algorithm efficiently finds the greatest common divisor (GCD).

GCD ensures numbers are coprime, a foundation of secure keys.

It underpins modular inverses and many encryption schemes.

Simple steps, profound impact on modern cryptography.



EULER'S TOTIENT FUNCTION AND THEOREM

TOTIENT FUNCTION Φ(N)

Counts the integers up to n that are coprime to n.

EULER'S THEOREM

For coprime a and n,
a^\(n) = 1 (mod n).

CRYPTOGRAPHIC IMPACT

Foundation for RSA and many other public key cryptosystems.

EXAMPLE

For n = 9

Numbers coprime to 9: 1, 2, 4, 5, 7, 8

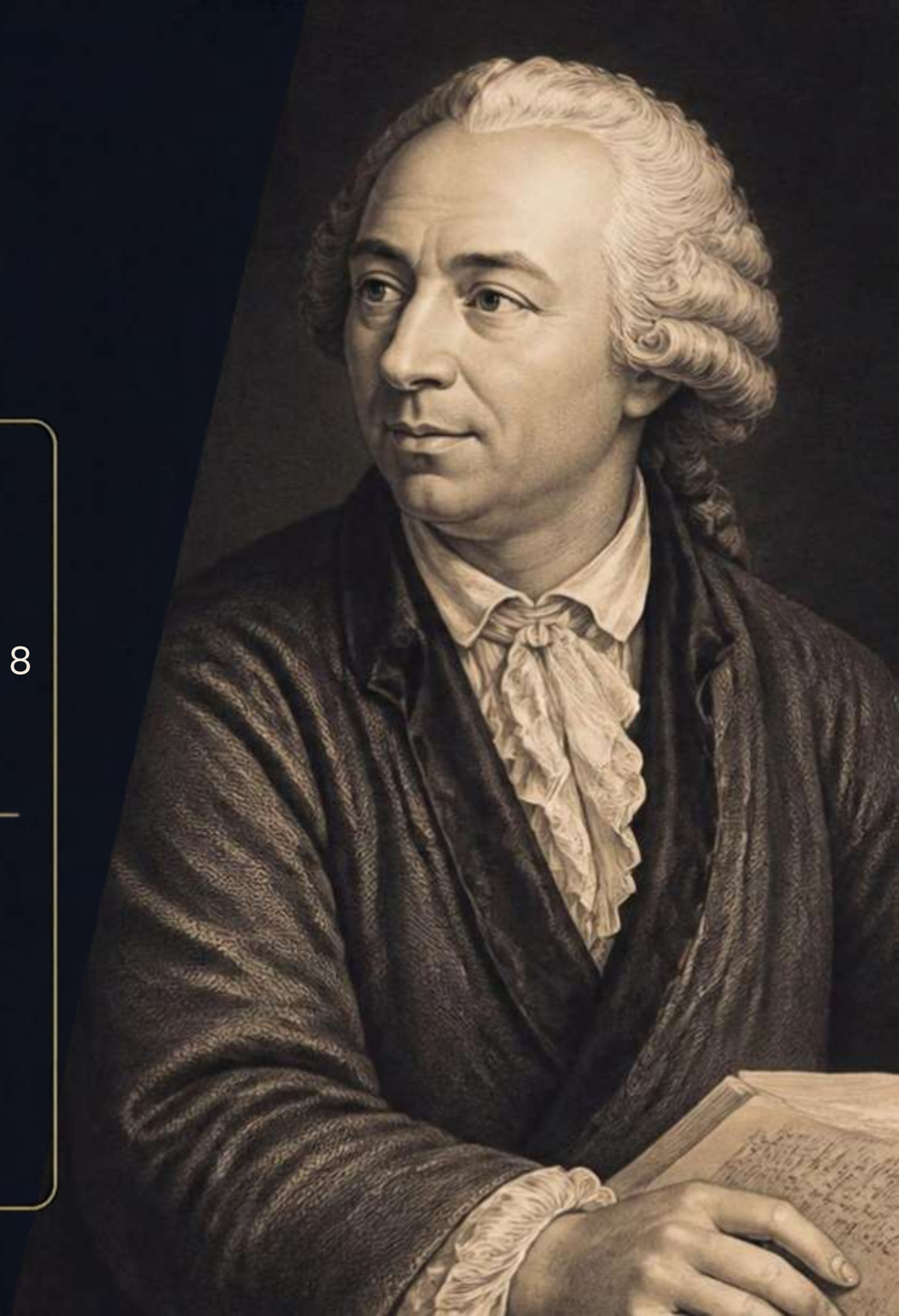
So, φ(9) = 6

Take a = 2

2^\(9) = 2^\(6 = 64

64 = 1 (mod 9)

Euler's theorem verified.



FROM THEORY TO PRACTICE

Number theory powers modern cryptography

KEY CRYPTOGRAPHIC ALGORITHMS	CLASSICAL NUMBER THEORY PROBLEM	CRYPTOGRAPHIC APPLICATION
RSA encryption and key generation	Factoring large integers	RSA encryption and key generation
Diffie-Hellman key exchange	Discrete logarithms in finite fields	Diffie-Hellman key exchange
ElGamal encryption	Discrete logarithms for encryption	ElGamal encryption



> Euclid's algorithms and Euler's theorems provide the structure that makes secure encryption possible.

KEY TAKEAWAYS

Core number theory powers algorithms and digital security

Euler’s totient function enables public key systems

Centuries of mathematics remain essential to modern protection

Mastering these ideas drives cryptographic innovation



THANK YOU FOR YOUR ATTENTION

Questions and discussion welcome

